Supplementary Information

Discovering adsorption rate laws from column data fails long before the rate constant is lost

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Supplementary Table 1: column, flow and physical parameters

Both adsorbents were solved in the same column geometry, particle size and superficial velocity. The step case also changes the bed porosity, particle density, heat of adsorption, feed and temperature, as recorded in erratum (b) of amendment A1. All values below are read from the code that built each solve, and the per-cell values from the ground-truth manifest.

The isothermal control uses the Langmuir column at Pe $\times$ 1 with no heat of adsorption. The feed-temperature van't Hoff factor is folded into b_{C0} and b_{H0} first, so the isotherm at T_{in} is the same as in the Langmuir cells and the bed stays at T_{in}.

Supplementary Note 1: methods, settings and known-answer tests

Observation models

Every method and every profile likelihood read data only through one function. O1 gives c, q and T at 20 equally spaced probe positions. O2 gives c and T at the same positions, with q hidden. O3 gives the outlet $c(t)$ and $T(t)$ only. All three take 200 times, evenly spaced over the solved horizon. The probe and time indices are rounded from a linear spacing of the stored 400 cells and 400 times.

Noise σ and isotherm error ε

Noise perturbs the observed channels only. Each observed channel gets independent Gaussian noise with standard deviation σ times that channel's range over the whole ground-truth field of the cell, not over the observed points. The levels were $\sigma \in \{0, 0.005, 0.02, 0.05\}$.

The isotherm error perturbs only the equilibrium the method is given. It does not touch the data. The method receives $q^{*}_{\mathrm{meas}}(c,T) = q^{*}(c,T)\,[1+\varepsilon\, g(c/c_{\mathrm{in}})]$, where $g(x) = \sum_{j=1}^{4} a_j \sin(j\pi x + \phi_j)$ divided by its root-mean-square on 2001 points of $[0,1]$, with a_j standard normal and ϕ_j uniform on $[0, 2\pi)$. The argument c/c_{in} is clipped to $[0,1]$. The function g is drawn once per (cell, replicate) and does not depend on σ or on the observation model. The levels were $\varepsilon \in \{0, 0.005, 0.02, 0.05\}$.

Every draw is seeded by the first 16 hexadecimal digits of a SHA-256 hash of its keys. The noise keys are (cell, replicate, σ, observation model); the isotherm-error keys are (cell, replicate). So the same inputs always give the same observation. A self-test checks that $\sigma = 0$ returns the stored field exactly, $\varepsilon = 0$ returns the true isotherm, the noise standard deviation is within 5 % of σ times the range, the isotherm error has root-mean-square ε to 10^{-3}, draws repeat, O2 hides q and O3 is outlet only.

Each cell had 60 replicates for every method run at the full count. Replicates 0 to 14 choose the best method of a slice; replicates 15 to 59 estimate every boundary.

Supplementary Table 1. Parameters of the two simulated columns. "Langmuir" is the default physics; "step" is water on a MOF-303-like framework with a Type V isotherm.

Quantity	Langmuir	Step
Geometry and flow		
Column length L	0.10	0.10
Inner diameter D_{in}	0.05	0.05
Particle diameter d_p	0.002	0.002
Superficial velocity v	0.10	0.10
Interstitial velocity $u = v/\varepsilon_t$	0.25	0.2857
Residence time L/u	0.40	0.35
Bed and solid		
Total porosity ε_t	0.40	0.35
Particle density ρ_p	1200	1100
Solid heat capacity C_{ps}	1000	1000
Axial thermal conductivity k_z	0.10	0.10
Gas and thermal boundary		
Gas density ρ_g	1.2	1.2
Gas heat capacity C_{pg}	1000	1000
Wall heat transfer coefficient h_w	10	10
Feed and wall temperature $T_{\mathrm{in}} = T_w$	298.0	303.15
Equilibrium		
Saturation capacity $q_{\max}$	20.0	25.0
Heat of adsorption ΔH	-50.0	-50.5
Cooperativity exponent n	1	4.0
Henry-site fraction f_H	1	0.06
Cluster affinity b_{C0}	1.720×10^{-9}	7.422×10^{-9}
Primary-site affinity b_{H0}	1.720×10^{-9}	3.711×10^{-10}
b_C at T_{in}	1.000	3.735
b_H at T_{in}	1.000	0.1868
Step position (cluster half saturation)	–	$P/P_0 = 0.15$ at 298.15
Feed concentration c_{in}	1.0	0.5043 (30 % RH)
$q^*(c_{\mathrm{in}}, T_{\mathrm{in}})$	10.00	21.90
$q^*(c_{\mathrm{in}}, T_{\mathrm{in}})/q_{\max}$	0.500	0.876
Step midpoint loading θ (L2 only)	–	0.3552
Kinetics, dispersion and design		
Stoichiometric time t_{stoich}	7200.4	31 050.4
$\mathrm{Da} = k\, t_{\mathrm{stoich}}$, 9 levels	10^{-1} to 10^{3}, half-decade spacing	
LDF coefficient k	1.389×10^{-5} to 0.1389	3.221×10^{-6} to 0.0322
Residence-time form kL/u	5.56×10^{-6} to 0.0556	1.13×10^{-6} to 0.0113
Default $D_L = 0.7D_m + 0.5\, d_p u$, $D_m = 2.5 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$	2.675×10^{-4}	3.032×10^{-4}
D_L multiplier ("Pe" level)	$\times 0.2$, $\times 1$, $\times 5$	
D_L at the three levels	5.35×10^{-5}, 2.675×10^{-4}, 1.338×10^{-3}	6.06×10^{-5}, 3.032×10
$\mathrm{Pe} = uL/D_L$ at the three levels	467, 93.5, 18.7	471, 94.2, 18.8
L2 rate drop and width	–	0.8, 0.05
Solves		
Grid cells N_z, stored times	400, 400	400, 400
Horizon (until outlet $c/c_{\mathrm{in}} > 0.95$)	4 or 8 t_{stoich}	4 or 8 t_{stoich}
Time integrator	BDF, rtol 10^{-6}, atol 10^{-10}, upwind advection	
Cells (L1, L2, isothermal control)	27, 0, 9	27, 27, 0

Known-answer system

Most method tests use one synthetic system that is never a grid cell: $\mathrm{d}q/\mathrm{d}t = k(q^*(c) - q)$ with $k = 0.02\,\mathrm{s}^{-1}$, a Langmuir $q^*(c) = 5 \cdot 3c/(1 + 3c)$, a logistic inlet ramp $c(t) = 1/(1 + \exp(-(t - 0.4\,t_{\mathrm{end}})/(0.05\,t_{\mathrm{end}})))$, $t_{\mathrm{end}} = 600\,\mathrm{s}$, constant $T = 298\,\mathrm{K}$, and 400 samples unless a test says otherwise. It is solved with rtol 10^{-10} and atol 10^{-12}.

Shared settings: library, derivative and selection floor

Lib-B is $\{1, c, q, T, cq, q^2, c^2, q^*_{\mathrm{meas}}\}$, with q^*_{meas} evaluated at the observed c and T. Lib-A is Lib-B without q^*_{meas}. Under L1 the true law in Lib-B is the support $\{q^*_{\mathrm{meas}}, q\}$. The strong-form target $\mathrm{d}q/\mathrm{d}t$ is a Savitzky–Golay derivative with window 21 and polynomial order 3 on the uniform time grid. Rows from all probes are stacked, so one law is fitted for the whole column.

Every sparse solver works on columns divided by their standard deviation (a constant column keeps scale 1). Coefficients are mapped back to the original units before scoring.

M2 and M2b: sparse regression in the weak and strong forms

The strong form (M2b) regresses the Savitzky–Golay derivative on Lib-B. The weak form (M2) multiplies the law by test functions $\phi(t) = ((t - a)(b - t))^8$ on windows of *support* samples, with a stride of support/4 (integer division), and integrates by parts with trapezoid weights, so the target needs no derivative of the data. Each ϕ is scaled to a maximum of 1. The support is swept over $\{31, 61, 121\}$ samples. For each support the system is fitted on the probes not held out and scored by $\|G\theta - b\|/\|b\|$ on the held-out probes, which are every fifth probe (indices 2, 7, 12 and 17 of 20). A support is admissible only if its training system has at least 3 equations per library term. The chosen support is then refitted on all probes. Both forms run all three solvers below.

Solvers. Sequentially thresholded least squares (STLSQ) uses ridge parameter 10^{-6} and drops a term when its standardised coefficient falls below 0.05 of the largest, for at most 20 iterations. The ensemble (M3) fits STLSQ with the same threshold on 100 bootstrap samples of 60 % of the rows, drawn with replacement from a fixed seed 0, keeps a term when it appears in at least 60 % of the fits, and reports its median coefficient. Exact best-subset selection (M4) enumerates every support of 1 to 4 terms, fits each by least squares, and picks the smallest BIC $= n \ln \max(\mathrm{RSS}/n,\ (2 \times 10^{-3}\,\mathrm{sd}(y))^2) + m \ln n$, with m the number of terms. M4 has no sign constraints.

Known-answer tests. On the known-answer system at zero noise, M4 recovers L1 with $|\hat{k} - k|/k < 0.02$, and STLSQ and the ensemble both recover L1. The weak form with M4 recovers L1 at $\sigma = 0$ and at $\sigma = 2\,\%$ with $|\hat{k} - k|/k < 0.05$, and at $\sigma = 2\,\%$ its k error is smaller than the strong form's. On a five-probe, 200-sample version, every solver recovers L1 in both forms through the grid's own pipeline. The residual floor 2×10^{-3} was the smallest of $\{1, 2, 5\} \times 10^{-3}$ that passed at 200 samples; the derivative's error there is 8×10^{-4} of $\mathrm{sd}(\mathrm{d}q/\mathrm{d}t)$.

M1b: uptake from the gas balance (O2)

On O2, q is recovered from the gas-phase balance of the solver, $\partial q/\partial t = (D_L c_{zz} - u c_z - c_t)/S$ with $u = v/\varepsilon_t$ and $S = (1 - \varepsilon_t)\rho_p/\varepsilon_t$, then integrated in time from $q = 0$ by the trapezoid rule. c_t is the Savitzky–Golay derivative above; c_z and c_{zz} are centred finite differences over the probe positions. The material and flow constants are taken as known. The two edge probes are dropped, so the O1 pipeline then runs on the inner 18 probes.

Known-answer tests. On a fresh solver run (default physics, $k = 0.002\,\mathrm{s}^{-1}$, $N_z = 200$, horizon 3 t_{stoich}, 300 times), the inversion recovers q to within 3 % of $\max q$ over the inner cells. Fed the true q, the O2 pipeline recovers L1 with $|\hat{k} - k|/k < 0.02$ (weak form, M4). With the inverted q it does not recover the law at this Da, which the pre-registration declared as a pilot fact.

M5: errors-in-variables best-subset (an oracle)

M5 runs M4's enumeration of 1 to 4 terms on Lib-B in the strong form, but fits each support by mixed least squares and total least squares. The constant column is treated as exact and projected out by least squares. The other columns and the target are scaled by their error standard deviations and solved by total least squares through the SVD. The selection uses the same floored BIC as M4. The error standard deviations are the declared ones: σ times the observed channel range for c, q and T; for q^*_{meas}, $\sqrt{(\varepsilon\, q^*_{\mathrm{meas}})^2 + (\partial q^*_{\mathrm{meas}}/\partial c\ \sigma_c)^2}$, with the derivative taken by a forward difference of step 10^{-6} times the range of c; first-order propagation for cq, q^2 and c^2; and for the target, the q-channel standard deviation times the norm of the Savitzky–Golay derivative coefficients (window 21, order 3). At $\sigma = 0$ the target standard deviation is set to 10^{-12}. Because M5 is given the true error levels, it takes no part in choosing the best method for H2a, H2b or the positive control, and is used in H2c only.

Known-answer tests. With $y = 2x + 1$ and noise of standard deviation 0.15 on x (20 000 points), ordinary least squares is attenuated by more than 0.2 in the slope and mixed least squares and total least squares recovers slope and intercept within 0.05. At near-zero declared error (10^{-6} of each column's standard deviation) on the known-answer system, M5 recovers L1 and selects the same support as M4. Through the grid wrapper at zero noise on the five-probe system, M5 recovers L1 and agrees with M4.

ODR-BINDy and WENDy were not run on the grid. The ODR-BINDy port recovered the exact 7-term support on its authors' Lorenz example in 1 of 5 noise seeds, with a mean relative coefficient error of 0.166 against about 2.2×10^{-3} reported, and is reported as "port, accuracy below the reference".

M6: SINDy-PI on the isothermal control

M6 is given no isotherm and no temperature. It runs only on the 9 isothermal-control cells, at every σ and at $\varepsilon = 0$, with 60 replicates, in the strong form. The library is $\{1, c, q, cq, q^2, c^2\}$, augmented with $\mathrm{d}q/\mathrm{d}t$, $c\,\mathrm{d}q/\mathrm{d}t$ and $q\,\mathrm{d}q/\mathrm{d}t$. Each of the three $\mathrm{d}q$ terms is tried as the left-hand side and fitted on the rest by M4's enumeration with up to 5 terms. The candidate with the smallest relative residual wins, and the relation is normalised so the $\mathrm{d}q/\mathrm{d}t$ coefficient is 1. Success requires the support $\{\mathrm{d}q, c\,\mathrm{d}q, c, q, cq\}$ exactly, with $c\,\mathrm{d}q > 0$, $c < 0$, $q > 0$ and $cq > 0$.

Known-answer test. On the known-answer system, M6 returns exactly the support above, and every coefficient is within 5 % of its true value ($c\,\mathrm{d}q = 3$, $c = -0.3$, $q = 0.02$, $cq = 0.06$). The errors measured on 2026-10-04 were 0.02 % and 0.07 % for b and k. Through the grid wrapper on the five-probe system, M6 succeeds.

M7: symbolic regression (PySR)

M7 runs PySR with binary operators $+$, $-$ and $\times$, no unary operators, 40 iterations, maximum expression size 12, a fixed random state equal to the replicate index, deterministic serial search, PySR's "best" model selection and a time limit of 1800 s per search. The features are Lib-B without

the constant, since PySR fits constants itself, and the target is the strong-form derivative. Feature names are passed as placeholders $f_0, f_1, \ldots$ and mapped back. The chosen expression is expanded by SymPy. A monomial that is a single feature to the first power counts under that feature; any other monomial is recorded as extra support, so a spurious term makes the fit fail. M7 runs on the subgrid of all Da, $\sigma \in \{0.5, 2\}\,\%$, $\varepsilon \in \{0, 2\}\,\%$, both isotherms, Pe $\times$ 1, O1, with 60 replicates.

Inclusion gate. Before any M7 search on the grid, PySR with these settings ran on the known-answer system with 0.5 % relative noise on q and on q^*_{meas}, for 3 seeds. The rule was to include M7 if L1 was recovered in at least 2 of 3 seeds, every search finished within 1800 s and peak memory stayed under 3 GB. All three seeds returned $k(q^*_{\mathrm{meas}} - q)$ with $\hat{k} = 0.019994$, 0.019992 and 0.019943 (relative errors 3.1×10^{-4}, 4.2×10^{-4} and 2.9×10^{-3}), in 35, 5 and 5 s, at a peak of 0.97 GB. M7 was included.

M8: Kolmogorov–Arnold network (not run on the grid)

M8 uses pykan 0.2.8 following Liu et al.'s procedure. It fits a KAN of width $[n, 1]$ (grid 5, spline order 3) on standardised raw variables $\{c, q, T, q^*_{\mathrm{meas}}\}$ with L-BFGS for 100 steps and sparsity penalty 10^{-3}, snaps every edge to $\{x, x^2, 0\}$, and reads the formula back in the original variables. It is not a reimplementation of KANDy.

Known-answer test and cost gate. M8 failed its known-answer test at zero noise in every configuration tried: a polynomial library at three sparsity penalties (10^{-3}, 10^{-2}, 10^{-1}), and raw variables on one and on three trajectories. It routed the law through q^2, used q alone, or returned $\{1, q^*_{\mathrm{meas}}, (q^*_{\mathrm{meas}})^2\}$. The test is kept as an expected failure. Its grid run was gated on one fit at the grid's size (20 probes, 200 times) taking at most 300 s. The measured fit took 1144 s, which projects to 68.7 h for the 216 subgrid fits, and returned only a constant. M8 was therefore not run on the grid.

M9: slow-manifold discovery

M9 regresses the state q, not its rate, on the q-free features $\{1, c, T, c^2, q^*_{\mathrm{meas}}\}$ by M4's enumeration with up to 3 terms and the same floored BIC. It runs on every non-isothermal L1 cell, O1, every (σ, ε), 60 replicates. Success is the support $\{q^*_{\mathrm{meas}}\}$ with coefficient 1 within 5 %.

Known-answer tests. On the known-answer system, M9 recovers $q = q^*_{\mathrm{meas}}$ at $k = 100\,\mathrm{s}^{-1}$ and correctly does not at $k = 0.002\,\mathrm{s}^{-1}$. Through the grid wrapper on the five-probe system at $k = 20\,\mathrm{s}^{-1}$, it succeeds.

M1: universal differential equation (demonstration)

M1 places a network in the uptake term of the column model. The network has two hidden layers of 32 tanh units. Its inputs are c/c_{in}, $q/q_{\max}$, $(T - T_{\mathrm{in}})/10\,\mathrm{K}$ and $q^*_{\mathrm{meas}}/q_{\max}$, and its output is multiplied by $q_{\max}/t_{\mathrm{stoich}}$, which needs no knowledge of k. Weights start as 0.3 times standard normal draws from a seed equal to the replicate index, and biases start at zero. The network is trained through a differentiable column ($N_z = 50$, Kvaerno5, PID step control with rtol 10^{-6} and atol 10^{-9}, initial step 10^{-3} s, at most 200 000 steps). The loss is the mean squared error of the outlet $c(t)$ and $T(t)$, each divided by its observed range. The optimiser is SciPy's L-BFGS-B with at most 50 iterations, 60 function evaluations and 7200 s, keeping the best parameters seen. If a

column solve exhausts its step budget after the first evaluation, training stops and keeps the best parameters (correction B88). The trained rate over all solved states is read out by M4 on Lib-B and by PySR with M7's settings; success on either counts. M1 ran once: Langmuir, Pe $\times$ 1, $\sigma = 2\,\%$, $\varepsilon = 0$, O3, the 9 Da levels, one replicate.

Pilot and tests. The JAX column matches the reference solver to better than 5×10^{-3} on the exit curve, and $N_z = 50$ differs from $N_z = 400$ by 0.55 %. One gradient with respect to a 2×32 network took 145 s, about 50 steps in 2 h, so M1 could not run at the declared scale and was reduced to the demonstration above. A smoke test on a fresh column ($k = 0.002\,\mathrm{s}^{-1}$) runs two L-BFGS steps and checks that the loss is finite and the readout returns; it does not test recovery. Three tests check the B88 rule: training keeps the best parameters when the solver fails, and a failure on the first evaluation is still an error. M1 has no known-answer recovery test.

Success metrics

L1 succeeds when the selected support is exactly $\{q^*_{\mathrm{meas}}, q\}$, the q^*_{meas} coefficient is positive, the q coefficient is negative, and their ratio is -1 within 10 %. The fitted rate constant is $\hat{k} = -\mathrm{coef}(q)$. For L2, with q^*_{meas} fixed, the implied rate $\hat{k}(q) = -(\mathrm{coef}(q) + \mathrm{coef}(cq)\,c + 2\,\mathrm{coef}(q^2)\,q)$ is averaged in 10 equal bins of $q/q_{\max}$; the state dependence is found when $\{q^*_{\mathrm{meas}}, q\}$ are selected, the Spearman correlation over the bins is below -0.8, and the top bin is at most 0.6 of the bottom bin. A fit that raised an error counts as a failure.

Profile-likelihood identifiability

Identifiability is judged with the true law known and with the same measured isotherm the methods receive. k is practically identifiable when the 95 % profile interval, the set where $2(\max \ell - \ell) \le 3.84$, lies within $[k/2, 2k]$. All profiles refine the same way: after the coarse grid, a pass of log-spaced points (41 on O1, 21 on O2 and O3) is placed between the outside neighbours of the current interval, for at most 6 passes, stopping when at least 5 points lie inside or the step ratio falls below 1.001. The reported ends are the outside neighbours, so the interval is conservative. As first run, the O1 profile used 21 points per pass; it was corrected to the pre-registered 41 before the corrected O1 rows were computed (Supplementary Note 2).

O1. The log-likelihood is the Gaussian sum over all 20 probes. For each probe, $q^*(t)$ is built by PCHIP interpolation of the observed c and T on a grid 4 times finer than the observations, and the LDF law is integrated by its exact update for a piecewise-linear q^*, vectorised over k. The coarse grid is $k =$ `logspace(-7, 1, 81)`, 10 points per decade, more than a decade beyond the smallest and largest k of the design. If the coarse likelihood peaks at either end of that grid, the profile raises an error, which counts as not identifiable. The noise standard deviation in the likelihood is $\max(\sigma, 10^{-4})$ times the range of the observed q. O1 identifiability runs on every non-isothermal L1 cell, every (σ, ε), 60 replicates.

O2 and O3. Each point of the profile is a full column solve at $N_z = 100$ with the measured isotherm, scored on the interior c and T probes (O2) or the outlet c and T (O3). The coarse grid is $k =$ `logspace(-6, 0, 25)`. The noise standard deviations are $\max(\sigma, 10^{-4})$ times the ground-truth range of c and of T. Each solve is capped at 600 s of wall time; a capped solve scores $-\infty$, and each row records the capped k values and whether one lies within one coarse step ($10^{6/24}$) of the interval.

These run on the subgrid of all Da, both isotherms, Pe × 1, L1, $\sigma \in \{0.5, 2\}\,\%$, $\varepsilon \in \{0, 2\}\,\%$, with 3 replicates (216 rows per observation model).

Known-answer tests. The exact O1 update agrees with the ODE-solver likelihood within 2×10^{-3} relative (plus 0.2) at $k \in \{0.005, 0.02, 0.05\}$. The O1 profile over 20 noisy probes ($\sigma = 2\,\%$) contains $k = 0.02$ and is identifiable. After A6, it contains k at 3.22×10^{-6}, 1.39×10^{-5} and $0.139\,\mathrm{s}^{-1}$, and it raises an error on a grid whose edge holds the peak. The single-probe profile on a fresh non-isothermal run ($k = 2 \times 10^{-4}\,\mathrm{s}^{-1}$, $\sigma = 0.5\,\%$) lies within 2 % of k. The outlet and probe profiles on a fresh low-Da run ($k = 2 \times 10^{-4}\,\mathrm{s}^{-1}$) contain k with hi/lo < 2. The solve cap scores a capped solve as $-\infty$, records it, and leaves an uncapped solve unchanged.

Supplementary Note 2: corrections made during the study

Every defect found in the design, code or text after it was written is recorded in a public ledger (`RETRACTIONS_P2.md`) with its cause, its fix and how it was found. Entries B76 and B79 to B84 were found before any grid result existed and reached no result; they corrected the analysis code, one method's inclusion gate, cost statements, citations and descriptions of the ground truth. B85, a provenance field missing from three results files, was found while the grid ran and affects no result. Four entries were found while the grid ran or after its first verdicts and changed what was computed. Each is described here with what it changed.

A5: a cap on each column solve (B86)

On O2 and O3, every point of the identifiability profile is a full column solve. With an isotherm error $\varepsilon > 0$, the measured isotherm evaluates its error term at c/c_{in} clipped to $[0, 1]$, and at large trial k the stiff solver kept crossing that kink: one O3 row made no progress for six days, and two O2 rows of the same cell took 39 and 42 hours. Amendment A5, made while the grid ran, capped each solve at 600 s and scored a capped solve as a failed solve. Every capped k is recorded with whether it lies next to the reported interval. In the finished O3 run, two rows had capped solves (six trial values of k each, all far above the interval, none adjacent to it). No reported interval depends on a capped solve.

A6: the range of the O1 profile (B87)

The pre-registered positive control requires discovery and identifiability to succeed at the lowest Da with no noise and no isotherm error; an isotherm that fails it is withheld from every verdict. As first run, the step isotherm failed: discovery succeeded in 135 of 135 estimation replicates and identifiability in none. The cause was the O1 profile's search range, 10^{-5} to $10\,\mathrm{s}^{-1}$, taken from a statement that holds for the Langmuir column only. The step cells at Da = 0.1 have $k = 3.2 \times 10^{-6}\,\mathrm{s}^{-1}$, below the range, so their likelihood peak was cut at the range's edge and every interval sat at $3.1\,k$. The stored fields obey the LDF law exactly there, and with a range that contains k the interval is $[1.0034, 1.0038]\,k$. Amendment A6, made after the first verdicts were seen, widened the range to 10^{-7} to $10\,\mathrm{s}^{-1}$ at the same density, made a peak on either edge an error, and recomputed all 51 840 O1 identifiability rows; the analysis was then rerun unchanged. A6 changes the range of a numerical search, chosen to contain every k of the design, and no criterion, threshold, estimand or verdict word.

Refinement points of the O1 profile (B89)

While drafting Supplementary Note 1, the O1 profile was found to refine with 21 points per pass, where the pre-registration declares 41 for this profile (21 is declared for the O2 and O3 profiles). Because the reported ends are the outside neighbours of the last points inside the interval, fewer points can widen an interval by up to one fine step, about 10 % instead of about 5 % for an interval of 0.6 decade. In the first run, 1 464 of the 13 849 non-identifiable O1 rows lay within 15 % of the $[k/2, 2k]$ limits with the other end inside. The A6 recomputation was 12 of 54 cells in when this was found; it was stopped, the refinement was set to 41, and the recomputation was restarted from the beginning, so every corrected O1 row uses the declared settings.

M1 kept its best parameters (B88)

Amendment A4 says the M1 demonstration keeps the best parameters seen during training. In four of its nine fits (Da = 31.6, 100, 316 and 1000), a solve of the differentiable column exhausted its step budget after one to two hours of training; only the time budget was handled, so the error ended the fit with no readout, and it was recorded as a failure. The training loop now stops on such a failure and keeps the best parameters seen; a failure on the first evaluation is still recorded as an error. The four fits were rerun with this code. Each stopped on a solver failure after 9 evaluations, with a training loss near 0.1, about a hundred times the loss the fits at Da $\leq$ 3.16 reached, and none recovered the law. M1 therefore succeeded in one of its nine fits (Da = 3.16, symbolic readout) in both runs. M1 enters no confirmatory test, so no verdict depends on it.

As-run and corrected verdicts

Both sets of verdicts are public (`results/p2_verdicts_asrun.json` and `results/p2_verdicts.json`). As first run, the Langmuir positive control passed (discovery and identifiability 135 of 135) and the step control failed, so the step isotherm was withheld from all 48 of its H2b slices. Among the 48 Langmuir slices, 20 had a verdict, and in all 20 discovery failed before identifiability; 28 had no boundary on the grid for discovery or identifiability. H2a and H2c had no verdict. The corrected verdicts are the primary result, because the first run did not compute the pre-registered estimand, the 95 % profile interval, for the affected cells.

	As first run	Corrected (primary)
Positive control, Langmuir	passed (135/135, 135/135)	passed (135/135, 135/135)
Positive control, step	failed (135/135, 0/135)	passed (135/135, 135/135)
O1 rows identifiable	37 991 of 51 840	40 891 of 51 840
H2b slices with a verdict	20 (Langmuir only)	30 (20 Langmuir, 10 step)
H2b overall	"*discovery fails before identifiability*"	"*discovery fails before identifiability*"
H2b gap, Langmuir (decade)	−1.60 to −3.43	−1.60 to −3.43
H2a	no verdict	no verdict
H2c	no verdict	no verdict

The corrections changed which isotherm could be judged, not what was found. The 2 900 O1 rows that became identifiable are step rows, nearly all in the cells A6 corrected; among the Langmuir rows, the 41-point refinement (B89) turned 24 rows identifiable and 17 non-identifiable, and the Langmuir H2b gaps moved by at most 0.11 decade (in one slice, where the identifiability boundary moved from Da $\approx$ 57 to 74), with every verdict unchanged.

Supplementary Note 3: exploratory analyses E2 and E3

Both analyses were declared in writing before they ran (`EXPLORATORY_P2.md`) and enter no confirmatory test.

E2: the Kolmogorov–Arnold network on its declared subgrid

M8 ran in exactly the configuration declared before the freeze (raw variables c, q, T, q^*; sparsity weight 10^{-3}; default training steps; pykan 0.2.8) on all nine Da levels, $\sigma \in \{0.5, 2\}$ %, $\varepsilon \in \{0, 2\}$ %, both isotherms, Pe ≈ 94, O1, three replicates: 216 fits. It recovered the law in none of them, at every Da (0 of 24 at each level) and in every condition (0 of 27 in each). In 62 fits the extracted law contained both q and q^*, always with additional terms, most often a constant.

Isotherm	σ, ε	fits	law recovered	q and q^* selected
Langmuir	0.5 %, 0	27	0	9
Langmuir	0.5 %, 2 %	27	0	8
Langmuir	2 %, 0	27	0	4
Langmuir	2 %, 2 %	27	0	3
step	0.5 %, 0	27	0	13
step	0.5 %, 2 %	27	0	12
step	2 %, 0	27	0	7
step	2 %, 2 %	27	0	6

One fit stopped with an error inside the library (a failed least-squares step in its spline refit, `kan/spline.py`, `curve2coef`); it is counted as a failure.

E3: precision against coverage of the identifiability criterion

The pre-registered criterion calls k practically identifiable when the 95 % profile interval lies inside $[k/2, 2k]$. It measures precision and accuracy within a factor of two, not whether the interval contains the true value. On O1, 40 891 of 51 840 rows met the criterion, but only 5 158 intervals (10 %) contained the true k; 87 % of the rows called identifiable had an interval that missed it. The same held on O2 (coverage 9 %; 157 of 216 rows identifiable) and, less strongly, on O3 (coverage 23 %; 150 of 216). The intervals are narrow because the likelihood treats the measured isotherm and the observed c and T that enter q^*_{meas} as exact, while the estimate carries small biases from them and from the time resolution.

Da	identifiable (of 5 760)	interval contains k	identifiable but misses k
0.1	5 760	425	92.6 %
0.316	5 760	454	92.1 %
1	5 760	646	88.8 %
3.16	5 760	1 042	81.9 %
10	5 760	1 511	73.8 %
31.6	5 558	709	87.2 %
100	3 308	205	93.8 %
316	1 467	159	89.6 %
1000	1 758	7	99.7 %